

3rd Sem.-B.Tech.

END-SEMESTER EXAMINATION, JULY 2026 (SUMMER SEMESTER)

Course Code- EAMTC04/ EPMTC301

Course Title- Mathematics for Machine Learning

Time- 03 Hours

Max. Marks-50

Note: - Attempt all five questions. Missing data/ information if any, may be suitably assumed & mentioned in the answer.

Q.No.	Attempt any 2 parts of the questions below	Marks	CO
1(a)	Find Kernel and Image of the linear transformation $T: R^4 \rightarrow R^3$ such that $T(x, y, z, w) = (x - z, y + w, x + y + z)$. Also, verify the Rank-Nullity Theorem.	05	CO1
1(b)	A mapping $T: R^3 \rightarrow R^3$ defined by $T(x, y, z) = (2x - y + z, x + 3y + 2z, 3x + 2y + z)$. Show that T is a linear mapping. Find $\text{Ker}T$ and the dimension of $\text{Ker}T$.	05	CO1
1(c)	Let V and W be two vector spaces in R^3 . Let $T: V \rightarrow W$ be a linear transformation defined by $T(x, y, z) = (x + y, y + z, x + z)$. Find the matrix representation of T with respect to the basis $X = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ in V and $Y = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}$ in W .	05	CO1
2(a)	Use the Gram-Schmidt process to convert the basis $\{(2,1,0), (1,2,1), (0,1,3)\}$ of the Euclidean space R^3 with standard inner product into an orthonormal basis.	05	CO2
2(b)	Find the eigenvalues and eigen vectors of $A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{bmatrix}$	05	CO2
2(c)	Reduce the quadratic form $Q = 3x^2 + 4y^2 + 5z^2 + 2xy + 2yz + 2zx$ to the normal form and check whether it is positive definite or positive semi-definite.	05	CO2
3(a)	Find the polar decomposition of $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.	05	CO3
3(b)	Find the pseudo-inverse of the matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}$.	05	CO3